

Proposed Methodology

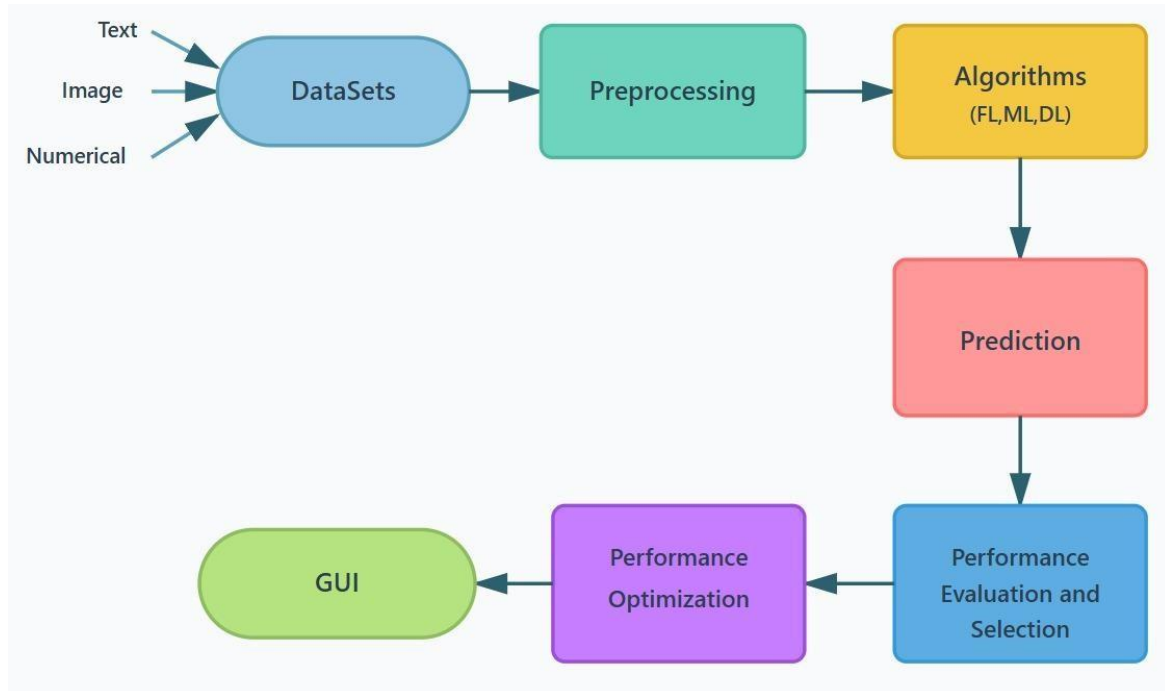


Fig 1. Proposed System Architecture

The system consists of a data processing structure that receives text, image and numerical information as input. Pre-processing of data is done to remove noise, modify missing values, standardize features with an intention of making them easier to analyze. The acquired processed data, in turn, is fed into the machine learning, deep learning, and federated learning algorithms to anticipate and categorize the data. The prediction results are evaluated using performance measures including accuracy, precision, recall and F1-score and the best performance model is selected. Further optimization techniques like transformer encoders are added so as to increase model efficiency and accuracy. Lastly, the findings are presented in the form of a graphical user interface (GUI), which allows the users to visualize predictions and system performance.